Land 1: Hearthvale Meadows

Body organization and homeostasis · OpenStax Ch. 1

Key ideas

Levels of organization

- From simplest to most complex: chemical, cellular, tissue, organ, organ system, organism.
- The cell is the smallest independently living unit; organelles are parts of cells.
- A tissue is a group of similar cells with a common function; humans have four types: epithelial, connective, muscle, nervous.
- An organ is made of two or more tissue types; an organ system is a group of organs working together.
- The human body has 11 organ systems.

Characteristics of life

- Organization: the body is highly ordered, and that order must be maintained.
- Metabolism is the sum of all chemical reactions: anabolism builds molecules and uses energy; catabolism breaks them down and releases energy.
- Responsiveness is sensing and reacting to change; movement occurs at the levels of the whole body, organs, and cells.
- Growth is an increase in size; development includes all lifetime changes, including differentiation of cells into specialized types.
- Reproduction occurs at the cellular level (cell division for growth and repair) and the organism level.

Requirements for life

- Oxygen is needed for aerobic respiration, which releases energy from nutrients and stores it in ATP.
- Nutrients supply energy and raw materials; water makes up about 60 to 70 percent of body weight and is the main solvent.
- Body temperature near 37 degrees C keeps enzyme driven reactions working; too cold slows them, too hot denatures proteins.
- Atmospheric pressure is required for breathing and for moving oxygen from the lungs into the blood.

Homeostasis and feedback

- Homeostasis is a dynamic equilibrium that keeps variables within a normal range around a set point.
- A feedback loop has a receptor (detects the change), a control center (compares with the set point and decides), and an effector (carries out the response).
- Negative feedback opposes the change and returns the variable toward the set point; examples are temperature, blood pressure, and blood glucose regulation.
- The hypothalamus is the control center for body temperature; sweating and vasodilation cool, shivering and vasoconstriction warm.
- Positive feedback amplifies the change until an end event stops it: childbirth (oxytocin and cervical stretch) and blood clotting (platelets recruiting platelets).
- Failure of homeostasis, such as uncontrolled blood glucose in diabetes, leads to disease.

Anatomical position and directional terms

- Anatomical position: standing upright, facing forward, feet flat and forward, arms at the sides, palms facing forward.
- Always use the patient's left and right, not the observer's.
- Supine is lying face up; prone is lying face down.
- Pairs: superior and inferior, anterior (ventral) and posterior (dorsal), medial and lateral, superficial and deep.
- Proximal and distal are used for limbs: proximal is closer to the trunk attachment, distal is farther away.

Body planes and sections

- Sagittal plane: divides left from right; midsagittal gives equal halves, parasagittal gives unequal parts.
- Frontal (coronal) plane: divides anterior from posterior.
- Transverse (horizontal) plane: divides superior from inferior; produces cross sections used in CT.
- A section is the flat surface produced by cutting along a plane.

Body regions and the abdomen

- Regional terms: cervical (neck), brachial (arm), antebrachial (forearm), carpal (wrist), femoral (thigh), crural (leg), sural (calf), popliteal (back of knee), antecubital (front of elbow), inguinal (groin), lumbar (lower back), gluteal (buttock).
- Four quadrants are formed by a midsagittal and a transverse line through the navel: RUQ, LUQ, RLQ, LLQ.
- Nine regions: top row right hypochondriac, epigastric, left hypochondriac; middle row right lumbar, umbilical, left lumbar; bottom row right iliac (inguinal), hypogastric (pubic), left iliac (inguinal).
- Examples: liver and gallbladder in the RUQ, spleen in the LUQ, appendix in the RLQ, sigmoid colon in the LLQ.

Body cavities and serous membranes

- Dorsal cavity: cranial cavity (brain) and vertebral canal (spinal cord).
- Ventral cavity: thoracic cavity above the diaphragm and abdominopelvic cavity below it.
- The thoracic cavity holds two pleural cavities (lungs) and the central mediastinum, which contains the heart in its pericardial cavity, plus the trachea, esophagus, and great vessels.
- Serous membranes have a parietal layer lining the cavity wall and a visceral layer covering the organ, with serous fluid between them to reduce friction.
- Pleura surrounds the lungs, pericardium surrounds the heart, and peritoneum lines the abdominopelvic cavity; inflammation causes pleurisy, pericarditis, or peritonitis.

Medical imaging

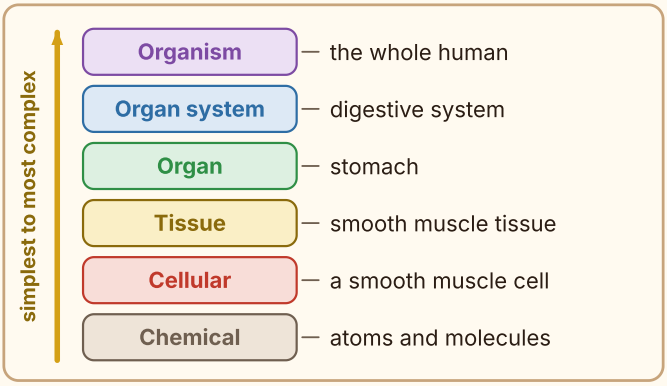
- X rays: dense tissue such as bone absorbs more radiation and appears white; uses ionizing radiation.
- CT: many X rays combined by computer into cross sectional (transverse) slices; fast and detailed.
- MRI: magnets and radio waves, no ionizing radiation; excellent for soft tissue such as brain and spinal cord.
- PET: radioactive tracer (often labeled glucose) shows metabolic activity of tissues.
- Ultrasound: high frequency sound waves, no ionizing radiation; preferred for monitoring pregnancy.

Villager wisdom

- My grandmother always said the body is built like a village: atoms build molecules, molecules build cells, cells make tissues, tissues make organs, and organs work together in systems.
- When the hearth gets too hot, you open a window. That is negative feedback: the response pushes things back toward the set point.
- Positive feedback is like a rolling snowball. It keeps growing until something outside stops it, as when the baby is born or the clot seals the cut.
- Always picture the traveler standing tall, facing you, palms forward. Every left, right, proximal and distal is named from that pose and from the patient's side.
- The lungs and heart sit in slippery sacs. Serous fluid between the two layers lets them glide; when the layers get inflamed, every breath or beat can hurt.

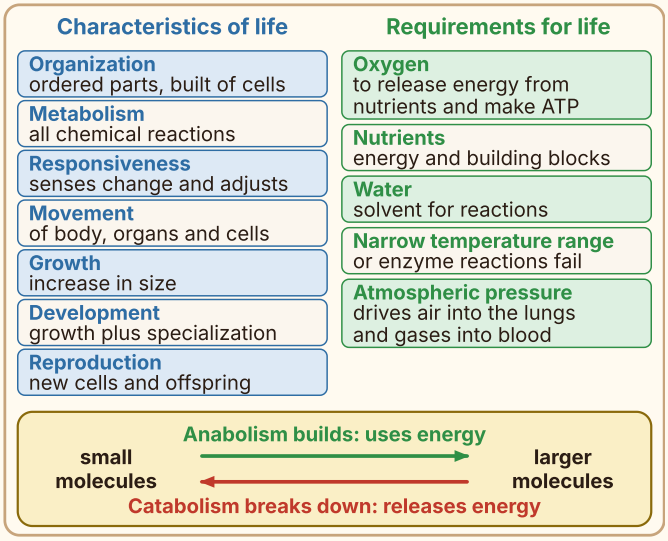
Diagrams

Levels of organization



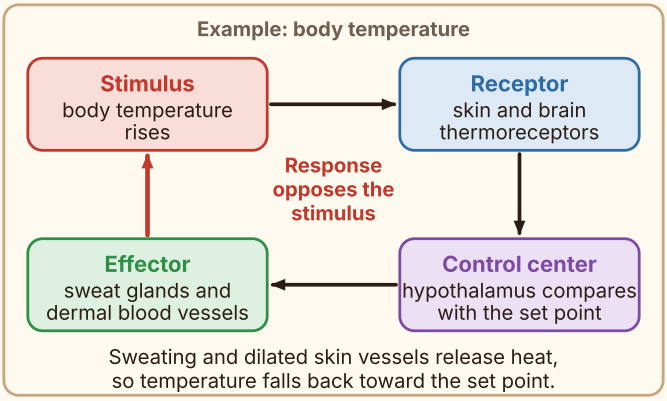
Each level is built from the one below it: chemicals form cells, cells form tissues, tissues form organs, organs form organ systems, and systems form the organism.

Characteristics and requirements of life



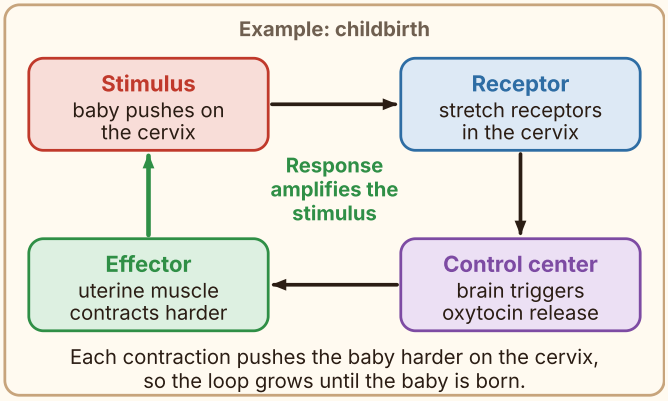
Living things share organization, metabolism, responsiveness, movement, growth, development and reproduction; survival needs oxygen, nutrients, water, a narrow temperature range and suitable atmospheric pressure.

Negative feedback loop



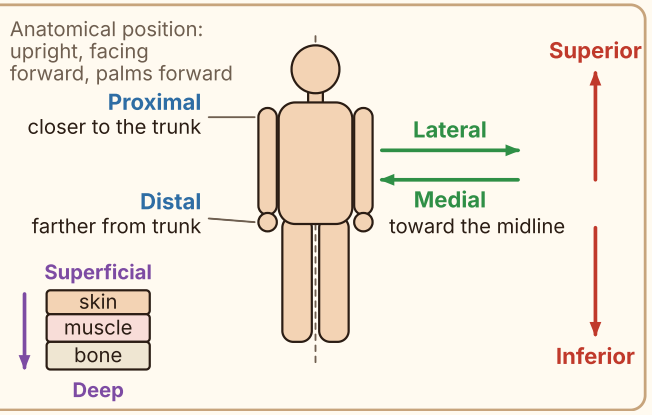
In negative feedback the effector response opposes the original change, returning the variable toward its set point; most homeostatic loops work this way.

Positive feedback loop



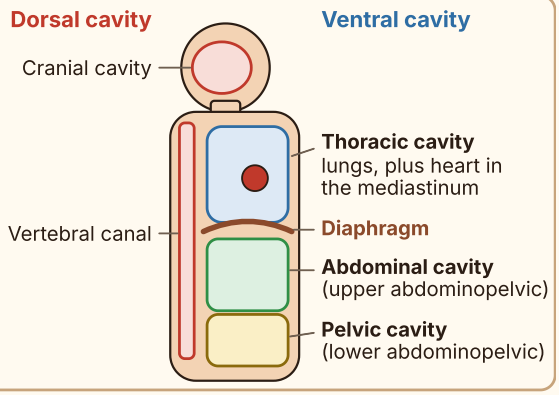
In positive feedback the response strengthens the original stimulus, so the change keeps growing until an outside event, such as birth, ends the loop.

Directional terms



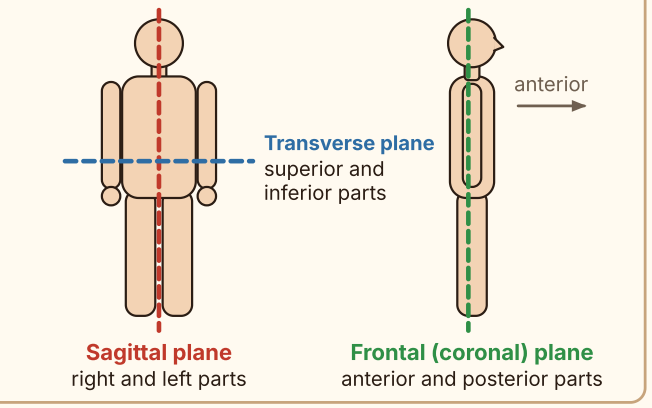
Directional terms compare two locations: the wrist is distal to the elbow, the nose is medial to the ears, and the skin is superficial to the muscles.

Body cavities



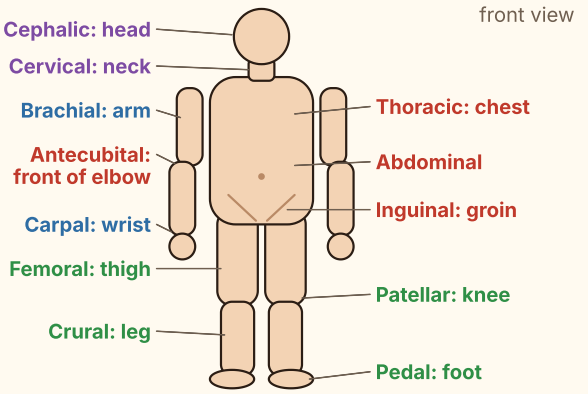
The dorsal cavity holds the brain and spinal cord; the ventral cavity is split by the diaphragm into the thoracic cavity above and the abdominopelvic cavity below.

Body planes



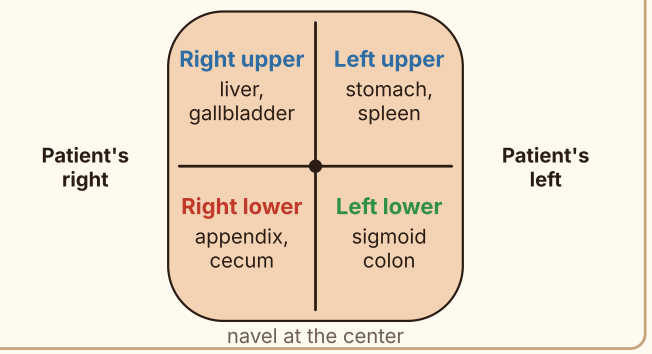
A sagittal plane divides right from left, a transverse plane divides superior from inferior, and a frontal (coronal) plane divides anterior from posterior.

Body regions



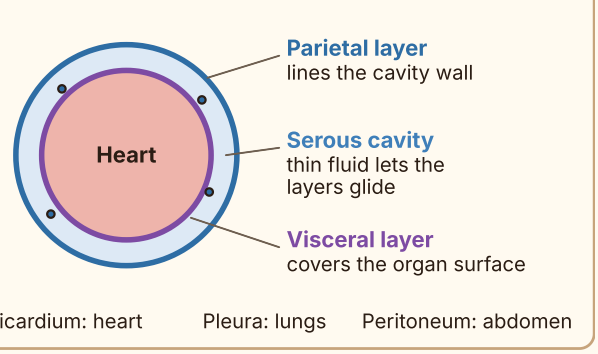
Regional terms name body areas: cervical is the neck, brachial the arm, antecubital the front of the elbow, inguinal the groin and femoral the thigh.

Abdominal quadrants



Lines through the navel make four quadrants named from the patient's point of view; the appendix lies in the right lower quadrant and the liver in the right upper.

Serous membranes



A serous membrane has a visceral layer on the organ and a parietal layer on the cavity wall; serous fluid between them lets the layers glide without friction.

Self check

1. **In a negative feedback loop, the response of the effector:**
    A. Amplifies the original stimulus
    B. Opposes the original stimulus
    C. Shuts off the sensor permanently
    D. Resets the set point higher
2. **A wound is on the back of the leg just above the heel. Which description is most accurate?**
    A. Posterior and distal leg
    B. Anterior and proximal leg
    C. Posterior and proximal leg
    D. Anterior and distal leg
3. **The stomach contains epithelial, muscle, connective, and nervous tissue working together. At what level of organization is the stomach?**
    A. Tissue
    B. Organ system
    C. Organism
    D. Organ
4. **What event normally ends the positive feedback loop of childbirth?**
    A. Oxytocin levels reach their set point
    B. Delivery of the baby ends the cervical stretch
    C. The hypothalamus detects low temperature
    D. The uterine muscle runs out of calcium
5. **Metabolism is best defined as:**
    A. The sum of all chemical reactions in the body
    B. The breakdown of food in the stomach only
    C. The building of new tissue during growth
    D. The removal of wastes by the kidneys
6. **Which part of metabolism builds larger molecules from smaller ones and requires an input of energy?**
    A. Catabolism
    B. Excretion
    C. Anabolism
    D. Respiration
7. **In anatomical position, the thumb is _____ to the little finger.**
    A. Medial
    B. Lateral
    C. Proximal
    D. Posterior

8. **A hiker in freezing weather develops severe hypothermia. Why does a core temperature far below normal become life threatening?**
    A. Enzyme driven reactions slow and organs fail
    B. Enzymes speed up and use up all the ATP
    C. Blood pressure rises until vessels rupture
    D. Cells instantly gain water and burst
9. **A patient's pain begins near the navel and then settles over the appendix. In which abdominal quadrant is the pain now?**
    A. Left upper quadrant
    B. Left lower quadrant
    C. Right lower quadrant
    D. Right upper quadrant
10. **An emergency note says a patient has a laceration in the right inguinal region. Where is the wound?**
    A. Groin
    B. Armpit
    C. Navel
    D. Lower back
11. **In anatomical position, the body stands upright and faces forward with:**
    A. Palms facing forward and feet flat
    B. Palms facing the sides of the thighs
    C. Arms raised above the head
    D. Palms facing backward
12. **A student describes the heart, blood vessels, and blood working together to transport nutrients and gases. Which level is she describing?**
    A. Organ
    B. Organ system
    C. Tissue
    D. Organism

Answer key

1. **B.** Opposes the original stimulus. Negative feedback reverses or opposes the change, pushing the variable back toward its set point. Amplifying the stimulus describes positive feedback, which drives a process further from its starting point.
2. **A.** Posterior and distal leg. The back of the body is posterior, and a point near the heel is farther from the hip attachment, so it is distal. Proximal would describe a spot near the knee instead.
3. **D.** Organ. An organ is a distinct structure made of two or more tissue types that performs specific functions. It is not an organ system, because the digestive system includes the stomach plus many other organs.
4. **B.** Delivery of the baby ends the cervical stretch. The stimulus driving the loop is stretch of the cervix; once the baby is delivered, the stretch stops and oxytocin release falls. Positive feedback loops do not aim for a set point, so oxytocin reaching a set point is not what ends it.
5. **A.** The sum of all chemical reactions in the body. Metabolism includes both anabolism (building larger molecules) and catabolism (breaking molecules down). Building new tissue is only the anabolic half, so it is too narrow to be the definition.

6. **C.** Anabolism. Anabolism links small molecules, such as amino acids into proteins, and consumes energy. Catabolism is the reverse: it breaks large molecules down and releases energy.
7. **B.** Lateral. With palms facing forward, the thumb lies farther from the midline than the little finger, so it is lateral. Medial would be correct only if the hand were rotated out of anatomical position.
8. **A.** Enzyme driven reactions slow and organs fail. Body chemistry depends on enzymes that work best near 37 degrees C; when cold, reactions slow until the heart and brain cannot function. Cold slows enzymes rather than speeding them up.
9. **C.** Right lower quadrant. The appendix usually sits in the right lower quadrant, so classic appendicitis pain migrates there. Quadrants are named from the patient's point of view, so the left lower quadrant is the opposite side.
10. **A.** Groin. The inguinal region is the groin, where the thigh meets the trunk. The armpit is the axillary region, a commonly confused term.
11. **A.** Palms facing forward and feet flat. Anatomical position places the arms at the sides with the palms forward, so the radius and ulna lie parallel. Palms resting against the thighs is how people usually stand, which makes it a tempting but incorrect choice.
12. **B.** Organ system. Several organs and tissues cooperating toward a shared role form an organ system, here the cardiovascular system. The heart alone would be an organ, but the description includes vessels and blood as well.

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